Task 2 Report: Next-Day AAPL Close Prediction

Healthcare + Data Science Project

March 27, 2026

Contents

1 Problem Statement **2**

2 Objective **2**

3 Dataset Loading and Preprocessing **2**

4 Model Training and Evaluation **2**

4.1 Primary Forecast Visualization 3

4.2 Additional Diagnostics 3

5 Results and Interpretation **3**

6 Conclusion **4**

1 Problem Statement

This task predicts the next-day closing price of AAPL using historical market data and engineered technical features, while avoiding time-series data leakage.

2 Objective

The modeling workflow is designed to:

- download and preprocess five years of daily stock data,
- create leakage-safe target and feature columns,
- train a regression model with chronological train-test split,
- evaluate prediction quality and visualize output behavior.

3 Dataset Loading and Preprocessing

Historical data was downloaded through `yfinance`. Preprocessing includes:

- target definition: `Target_Next_Close = Close.shift(-1)`,
- feature engineering: `SMA_10`, `SMA_50`, `Daily_Return`,
- null-row removal after rolling and shifting operations.

4 Model Training and Evaluation

Random Forest Regressor was trained with:

- chronological split (first 80% train, last 20% test),
- feature scaling fit only on training data,
- evaluation metrics: MAE and RMSE.

4.1 Primary Forecast Visualization

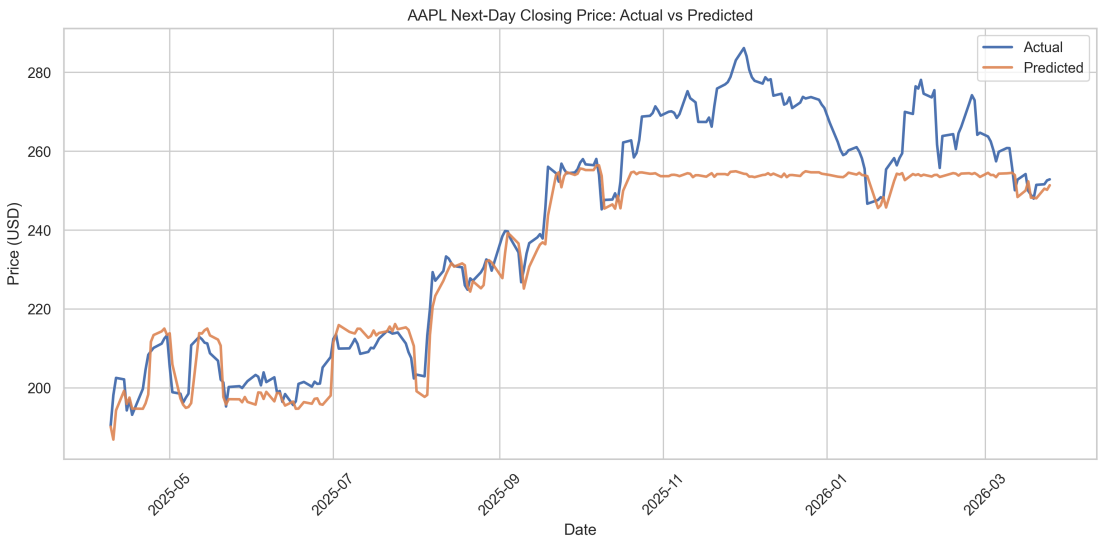

Figure 1: Actual vs predicted next-day closing prices for test period.

4.2 Additional Diagnostics

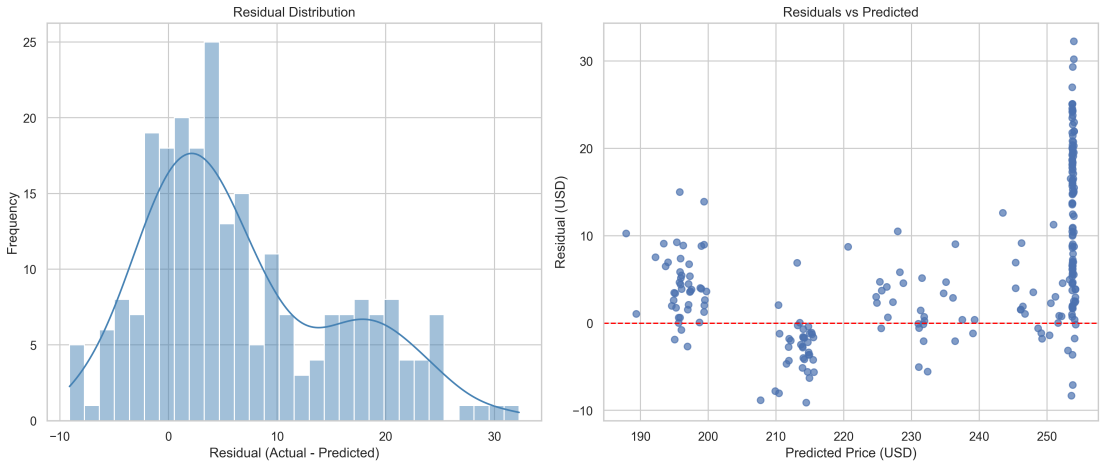

Figure 2: Residual diagnostics for error behavior and distribution.

5 Results and Interpretation

- The pipeline is technically robust with leakage prevention and clear chronological validation.
- MAE/RMSE should be interpreted against a naive baseline for practical signal quality.
- Next-day stock close prediction remains difficult due to high short-term market noise.
- This setup forms a solid baseline for future feature enrichment and model comparison.

6 Conclusion

The task demonstrates an end-to-end time-series modeling workflow with proper safeguards and explainable outputs. Improvements should focus on stronger feature design, baseline benchmarking, and probabilistic uncertainty-aware evaluation.